



A commentary by David N. Bernstein, MD, PhD, MBA, MEI, is linked to the online version of this article.

Behavioral Intervention to Foster Healthy Lifestyle Physical Activity After Complex Lumbar Surgery

A Randomized Controlled Trial

Carol A. Mancuso, MD, Roland Duculan, MD, Manuela C. Lafage, BSN, Frank P. Cammisia, MD, Andrew A. Sama, MD, Alexander P. Hughes, MD, Darren R. Lebl, MD, and Federico P. Girardi, MD

Investigation performed at the Hospital for Special Surgery, New York, NY

Background: Persistent sedentary lifestyles are prevalent after complex lumbar surgery. Inactivity often is due to habit, is not necessary, and results in both adverse general health and adverse spine health. We tested a multicomponent behavioral intervention administered in surgical practices to increase lifestyle walking after recuperation from surgery involving ≥ 3 lumbar levels or fusion. This analysis reports the main objective of measuring change in physical activity; subsequent analysis will address the additional objective of measuring change in disability due to lumbar pain.

Methods: At 3 months postoperatively, patients were interviewed in person, completed the self-reported Paffenbarger Physical Activity and Exercise Index measuring energy expenditure in kcal/week from walking and total activity, and were randomized to the control or a behavioral intervention. Patients in the intervention group made a contract to increase walking and received a pedometer and information about activity benefits, followed by telephone contacts over 1 year to reinforce the intervention and assess activity change. Controls received only information about activity benefits and fewer telephone contacts to assess activity change. The primary outcome was the difference in within-patient change in kcal/week from walking (as measured by the Paffenbarger Index) from enrollment to 4 months later. Additional assessments and outcomes included changes in the total Paffenbarger Index from enrollment to 4 and 12 months and changes in the walking domain to 12 months.

Results: Among 250 patients (mean age, 63 years; 53% men; 96% White patients and 6% Latino patients), 123 received the intervention and 127 were controls. The mean energy expenditure from walking increased from enrollment to 4 months for patients in the intervention group (1,437 to 2,582 kcal/week; $p < 0.0001$) and for controls (1,320 to 1,870 kcal/week; $p < 0.0001$), with a greater difference for the intervention group (1,165 compared with 600 kcal/week; $p = 0.03$). At 12 months, the effectiveness of the intervention was sustained (estimated effect, 588 kcal/week [95% confidence interval (CI), 119 to 1,056 kcal/week]; $p = 0.01$) when accounting for new events that potentially affect activity, such as another lumbar surgery (estimate, $-1,396$ kcal/week [95% CI, $-2,116$ to -676 kcal/week]; $p = 0.0002$) and new arthroplasty or foot or ankle surgery (estimate, -701 kcal/week [95% CI, $-1,212$ to -189 kcal/week]; $p = 0.007$). Similar results were obtained for changes in total activity. Retention in this study was 92%; 72% of all patients reported that being in the study did not affect spine symptoms, and 28% reported that being in the study made symptoms better. No adverse events were associated with the trial.

Conclusions: A behavioral intervention was effective in increasing lifestyle walking after recuperation from complex lumbar surgery. These findings support integrating counseling about physical activity into routine postoperative care in surgical practices.

Level of Evidence: Therapeutic Level I. See Instructions for Authors for a complete description of levels of evidence.

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Physical activity is essential for both general health and spine health¹. In addition to fostering muscle strength and bone formation, prudent activity promotes proper alignment and weight maintenance, which are necessary for the postoperative lumbar spine.

Painful lumbar disorders are known to lead to avoiding physical activity and adopting lifestyles that are sedentary (i.e., mostly sitting or lying down)²⁻⁴. Interestingly, although acknowledging the benefits of activity, many patients remain inactive postoperatively, with activity levels below population norms^{3,5-7}. Covariates and reasons for persistent inactivity include residual pain, fear of injury, more complex surgery, and depressive symptoms^{2,3,6-8}. These factors are superimposed on a background of general reasons for inactivity, including age, comorbidity, body mass index, habit, and life circumstances that encroach on access to or time for activity^{2,3,7-9}.

Behavioral interventions are known to effectively increase physical activity in patients with musculoskeletal conditions¹⁰. These interventions, based on precepts such as the social cognitive theory, involve providing information about the benefits of physical activity, increasing self-efficacy (i.e., the confidence or belief in the ability to succeed in a situation), setting activity goals, providing ongoing approbation and positive feedback from study personnel for healthy behaviors, and physiology feedback of success as would be provided by a motion-sensing device such as a pedometer¹¹⁻¹⁴. Outcomes are often self-reported as time spent in activities or energy expenditure in kilocalories per week. It is not known whether such interventions would be effective in patients after complex lumbar surgery.

To address this question, we conducted a trial to foster physical activity, taking into account several issues. First, patients need approbation from surgeons that physical activity is good for the spine; therefore, we conducted the trial in spine practices after surgeons affirmed that it was safe for each patient to increase physical activity. Second, introducing an intervention postoperatively, rather than preoperatively, would be more logical, as patients would not be limited by the most severe pain or distracted by concerns about surgery. Third, the activity should be something that everyone could do safely, such as increased lifestyle walking integrated into daily routines. Fourth, on the basis of prior studies, we selected a 4-month outcome, given that it is not possible to continually increase activity indefinitely, to assess the effectiveness of implementing the intervention and a 12-month outcome to assess the effectiveness of maintaining the intervention^{8,15,16}. Fifth, to account for the role of social interaction in fostering behavior change, the comparator group should be an attention control group and receive a prespecified protocol of interactions with study personnel.

The study question was whether a behavioral intervention instituted during a randomized trial after complex lumbar surgery would be effective in improving self-reported physical activity. The hypothesis was that a multicomponent psychosocial behavioral intervention would be effective in increas-

ing the within-patient change in kcal/week from walking at 4 months after enrollment, which was the primary outcome. Additional outcomes were the changes in kcal/week from all activities at 4 and 12 months.

Materials and Methods

This trial was approved by our institutional review board, and all patients provided written informed consent. Consecutive patients of 5 participating surgeons were screened, enrolled, and interviewed in person during routine 3-month postoperative visits. Patients were eligible if they had had complex lumbar surgery based on the validated Surgical Invasiveness Index, which considers the number of vertebral levels involved, decompression, fusion with or without instrumentation, and approach¹⁷; were ≥ 18 years of age; and spoke English; and if their surgeon confirmed that it was safe to increase lifestyle walking. Patients were excluded if they reported cardiopulmonary symptoms on a physical activity preparticipation screening questionnaire¹⁸. The study was registered at ClinicalTrials.gov (NCT02611427). This analysis reports the main objective of measuring change in physical activity; subsequent analysis will address the additional objective of measuring change in disability due to lumbar pain.

Information from Patients at Enrollment

Patients self-reported their age, sex, race, and ethnicity. Patients completed the Paffenbarger Physical Activity and Exercise Index, an epidemiological measure of physical activity in 3 domains¹⁹. Patients were asked how many blocks they walked daily, how many stairs they climbed daily, and the frequency and duration of sports and exercise during the past week. If walking was reported in miles, a conversion of 1 mile = 20 blocks was made. Weekly energy expenditure in kcal/week was calculated on the basis of a standard compendium that assigns kilocalories to different activities and sports and considers the amount of the activity and sport performed²⁰. For example, walking 20 blocks daily corresponds to 1,120 kcal/week (i.e., 20 blocks/day $\times$ 8 kcal/block $\times$ 7 days/week). Coupling the Paffenbarger Index and the compendium provided a comprehensive self-reported account of baseline and within-patient change in energy expenditure. The kcal/week were calculated for each domain and were summed to represent total physical activity. Patients also reported their previous activity behaviors (i.e., participation in sports or exercise 10 years ago and when they were teenagers or young adults).

Using validated surveys, patients were asked about medical conditions (Charlson Comorbidity Index), depressive symptoms (Patient Health Questionnaire-8 [PHQ-8]), pain-related disability (Oswestry Disability Index), and fear of increasing back pain (Fear-Avoidance Beliefs Questionnaire)²¹⁻²⁴. After completing all surveys, patients were randomized 1:1 to either the control or intervention group according to a blocked sequence, unknown to the recruiting investigators, of initially 10 and then 20 patients. Randomization was not stratified for any variable, and assignments were provided to recruiting investigators at the time of each enrollment.

The study entailed in-depth, in-person interviews at enrollment, followed by telephone contacts throughout the study period.

Intervention Group

Patients in the intervention group received a multicomponent behavioral intervention based on social cognitive theory¹¹. First, patients were given the 2008 Physical Activity Guidelines for Americans, and sections on increasing lifestyle walking were emphasized¹. Second, to increase self-efficacy or self-confidence, patients described prior challenging life events that they successfully overcame. Third, patients made a contract to increase walking that they were confident they could fulfill (e.g., how many more blocks/day, incorporating more walking into daily routines). Patients could also report walking in terms of miles (including on a treadmill), and this was then converted as above. Patients in the intervention group were also given a DIGI WALKER pedometer (Yamax), which has been validated by the manufacturer during treadmill testing and free-living walking^{25,26}. For patients in our trial, the pedometer was calibrated to individual stride length and was used to provide physiological evidence of successfully walking more.

Patients in the intervention group were contacted by telephone weekly, and then biweekly, for 5 months and then monthly up to 14 months to receive approbation or positive feedback from study personnel and to report interval outcome data (each contact was approximately 15 minutes). Three major time points were defined: (1) the 4-month outcome (i.e., during

the 4 to 6-month trial follow-up), (2) 8-month assessment (i.e., during the 7 to 10-month follow-up), and (3) 12-month outcome (i.e., after 11 months and up to 14 months). Patients in the intervention group could receive a total of 11 contacts within the 4-month period and 22 contacts within the 12-month period.

Attention Control Group

Patients in the control group received the 2008 Physical Activity Guidelines for Americans, and study personnel emphasized increasing lifestyle walking¹. Patients were contacted by telephone monthly up to 14 months. Three major time points were similarly defined around 4, 8, and 12-month assessments and outcomes. Patients in the control group could receive a total of 4 contacts within the 4-month period and 14 contacts within the 12-month period.

Definition of Outcomes

The primary outcome was the difference between groups in the within-patient change, from enrollment to 4 months, in energy expenditure (kcal/week) from blocks walked, as measured by the Paffenbarger Index. A secondary outcome was the difference between groups in the within-patient change in total kcal/week (sum of all Paffenbarger Index domains) from enrollment to 4 months. Additional secondary outcomes were the differences between groups in the within-patient change in kcal/week from blocks walked and in total kcal/week at 12 months. A validated self-reported outcome was used instead of a wearable device, which often is not systematically worn for the long term.

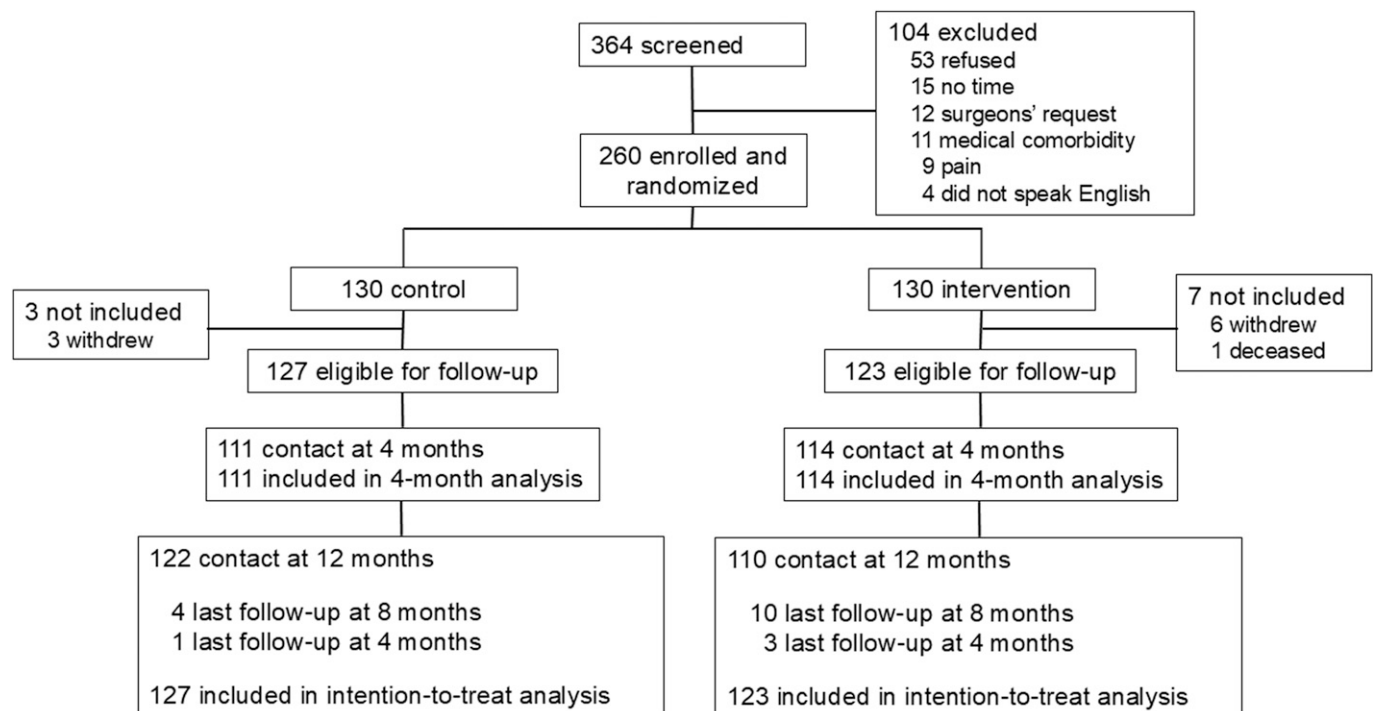


Fig. 1
Flow diagram of patient participation.

TABLE I Enrollment Characteristics According to Randomization Group

Variable	Control* (N = 127)	Intervention* (N = 123)	P Value
Age (yr)	63 ± 11 (34 to 83)	64 ± 12 (27 to 81)	0.57
Male	50%	55%	0.44
Race or ethnicity			
White	95%	98%	0.57
Latino	6%	7%	0.74
Body mass index (kg/m ²)	28 ± 5	29 ± 6	0.19
Major comorbidity†	42%	33%	0.17
Patient Health Questionnaire (PHQ-8), depressive symptoms score‡	5 ± 5	5 ± 5	0.59
Fear-Avoidance Beliefs Questionnaire score§	11 ± 8	11 ± 8	0.96
Participated in sports and/or exercise in the past 10 years	75%	85%	0.05
Participated in sports and/or exercise in youth	65%	70%	0.36
Oswestry Disability Index score#	34 ± 18	34 ± 16	0.90
Diagnosis			0.09
Degenerative disc disease	6%	4%	
Stenosis	41%	42%	
Degenerative spondylolisthesis	35%	32%	
Lumbar scoliosis	17%	17%	
Other**	1%	5%	
Surgical Invasiveness Index score††	11 ± 6	13 ± 7	0.06
Underwent operation on ≥3 levels	45%	50%	0.34
Had fusion	89%	93%	0.31
Had instrumentation	86%	92%	0.13

*The values are given either as the mean and the standard deviation, with or without the range in parentheses, or as the percentage of patients.

†This category is based on the Charlson Comorbidity Index. ‡The possible range for this category is 0 to 24; a higher score indicates more depressive symptoms. §The possible range for this category is 0 to 24; a higher score indicates more fear-avoidance beliefs. #The possible range for this category is 0 to 100; a higher score indicates more disability. **This subcategory includes nonunion, pain due to the implant, and spondylolytic spondylolisthesis. ††The maximum score is 50; a higher score indicates more complex surgery.

Blinding

Patients were blinded to the randomization group. In order to enhance rapport and foster using pedometers and fulfillment of walking contracts, the same investigators enrolled patients, conducted telephone interviews, and administered follow-up questionnaires.

Clinical Information

Medical records were reviewed for operative details to classify surgical complexity¹⁷.

Information from Patients During Follow-ups

During each telephone contact, patients in both groups were asked about untoward events from increased walking (e.g., back pain, muscle and/or joint symptoms, new or worsening medical conditions). Both groups completed the Paffenbarger Index again at 4, 8, and 12 months. Suggestions were offered for indoor walking if weather hindered outdoor activity. Patients also were asked about any new clinical or personal life events that impeded activity efforts. Responses were grouped into spine-related events (e.g., another lumbar surgery), lower-extremity events

(e.g., hip and knee arthroplasty, foot and ankle surgery), and other musculoskeletal, medical, or personal events. At the final follow-up, patients were asked how participation in this trial affected the spine symptoms (i.e., no effect, symptoms were better, or symptoms were worse).

Sample Size

The sample size was based on the 4-month outcome. In prior longitudinal studies, standard deviations for weekly energy expenditure ranged from 350 to 900 kcal/week and an increase in 336 kcal/week (equivalent to 6 blocks/day) corresponded to decreased cardiovascular risk^{15,27,28}. Using a standard deviation of 800 kcal/week, a goal of detecting a difference of 250 kcal/week (equivalent to 4 blocks/day), an alpha of 0.05, and a power of 90%, we required 106 patients per group. Assuming attrition of 15%, 122 patients per group (244 total) were required.

Statistical Analysis

Energy expenditure in kcal/week was calculated for walking, stairs, and exercise and sports (i.e., the Paffenbarger Index

TABLE II Paffenbarger Physical Activity and Exercise Index: Baseline and Longitudinal Values According to the Randomization Group

Assessment	Control* (kcal/week)	Intervention* (kcal/week)	Difference Between Groups* (kcal/week)	P Value
At enrollment†				
Walking domain	1,320	1,437	117 (–445 to 211)	0.48
Total of all domains	1,769	1,808	39 (–475 to 397)	0.86
At 4 months‡				
Walking domain	1,870	2,582	712 (166 to 1,258)	0.01
Change from enrollment	600§	1,165§	565 (59 to 1,071)	0.03
Total of all domains	2,756	3,539	783 (54 to 1,512)	0.04
Change from enrollment	1,092§	1,750§	658 (45 to 1,272)	0.04
At 8 months#				
Walking domain	2,094	2,542	448 (–122 to 1,018)	0.12
Change from enrollment	782	1,076	294 (–256 to 844)	0.29
Total of all domains	2,839	3,594	755 (–40 to 1,550)	0.06
Change from enrollment	1,199	1,781	582 (–168 to 1,332)	0.13
At 12 months†				
Walking domain	1,662	2,220	558 (69 to 1,048)	0.03
Change from enrollment	341**	783§	441 (–40 to 923)	0.07
Total of all domains	2,559	2,985	426 (–215 to 1,068)	0.19
Change from enrollment	791§	1,178§	387 (–175 to 949)	0.18

*The values are given as the mean, with or without the 95% CI in parentheses. †There were 127 patients in the control group and 123 patients in the intervention group. ‡There were 111 patients in the control group and 114 patients in the intervention group. §P ≤ 0.0001. #There were 89 patients in the control group and 104 patients in the intervention group. **P ≤ 0.04.

domains). Demographic and clinical characteristics and Paffenbarger Index scores at enrollment were compared between randomization groups and according to surgeon. The numbers of telephone contacts per patient were recorded. As noted above, 3 time points were defined (4, 8, and 12 months from the date of enrollment in the trial). The within-patient change

from enrollment to 4, 8, and 12 months in the Paffenbarger Index walking and total values were compared with paired t tests, and differences were compared between groups with independent t tests. The main outcome was the within-patient change in kcal/week for the walking domain at 4 months. Secondary outcomes were similar comparisons for the 4-month

TABLE III Linear Regression Models for Paffenbarger Physical Activity and Exercise Index (Kcal/Week) at 4 and 12 Months

Variable	Estimated Effect*	P Value
4-month walking domain		
Intervention group	565 (59 to 1,071)	0.03
4-month total physical activity		
Intervention group	658 (43 to 1,274)	0.04
12-month walking domain†		
Intervention group	588 (119 to 1,056)	0.01
Another lumbar surgery	–1,396 (–2,116 to –676)	0.0002
Arthroplasty or foot or ankle surgery	–701 (–1,212 to –189)	0.007
12-month total physical activity†		
Intervention group	578 (39 to 1,118)	0.04
Another lumbar surgery	–1,540 (–2,371 to –710)	0.0003
Arthroplasty/foot or ankle surgery	–1,181 (–1,771 to –591)	0.0001

*The values are given as the estimate, with the 95% CI in parentheses. †Determined with use of multivariable analysis.

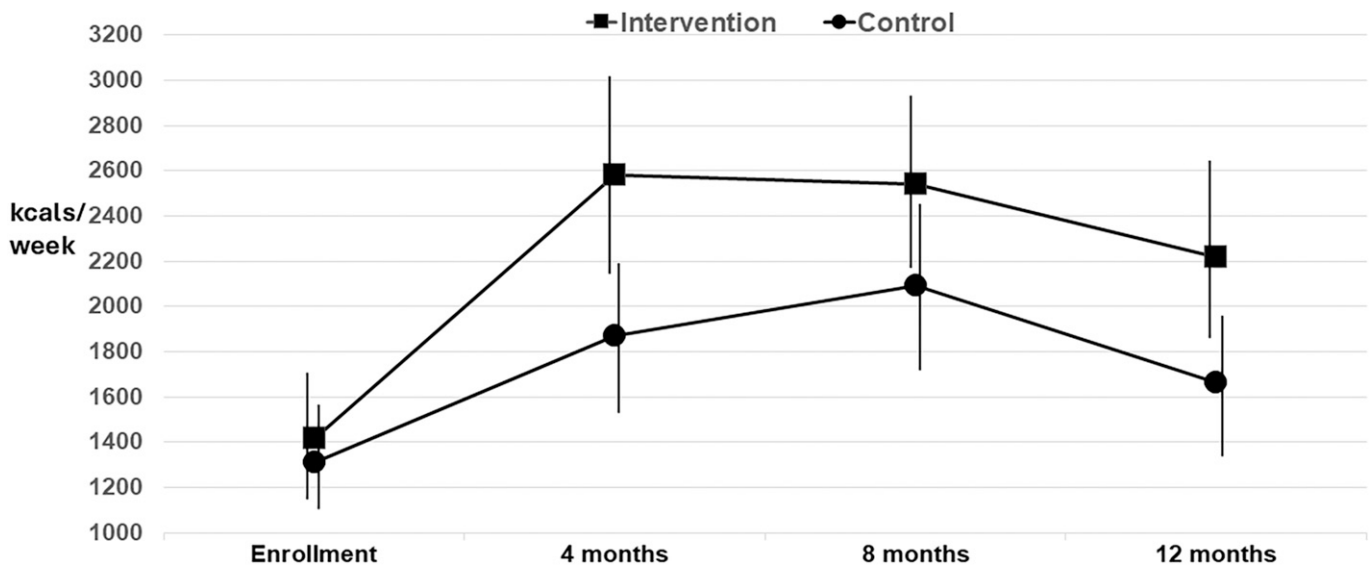


Fig. 2

Mean energy expenditure in kcal/week with 95% CIs for the walking domain at follow-ups according to the randomization group.

total and 12-month walking and total values. Intention-to-treat analyses were performed for the 12-month outcomes, with the last value carried forward for patients without 12-month data. The 12-month analyses were performed by multivariable modeling, including clinical or personal life events (i.e., lumbar and lower-extremity surgeries since enrollment) for which patients reported impeded activity efforts.

Results

From November 2015 to February 2018, 364 consecutive patients were screened; 260 were enrolled, and 130 were randomized to the control and 130 were randomized to the intervention (Fig. 1). Nine patients withdrew (due to unhappiness with the surgical outcome, no time, pressing family matters), and 1 patient died (not related to the trial). Thus, 127 patients in the control group and 123 patients in the intervention group were eligible for follow-ups. Patients who did or did not withdraw were similar in Paffenbarger Index values and demographic and clinical characteristics ($p > 0.05$ for all comparisons).

The mean age (and standard deviation) of the entire sample was 63 ± 11 years, 132 patients (53%) were men, 241 (96%) were White, 15 (6%) were Latino, and the mean time since surgery was 2.9 ± 0.4 months. Randomization was successful, with similar demographic and clinical characteristics (Table I) and Paffenbarger Index values (Table II) between groups and according to surgeon.

Primary Outcome at 4 Months

For the 123 patients in the intervention group, 114 completed the Paffenbarger Index at a mean of 4.2 ± 0.5 months; of these patients, 89 (78%) had ≥ 6 of the possible 11 telephone contacts. For the 127 patients in the control group, 111 completed the Paffenbarger Index at a mean of 4.2 ± 0.6 months; of these patients, 95 (86%) had ≥ 3 of the possible 4 telephone contacts.

The mean kcal/week for the walking domain had increased at 4 months for each group ($p < 0.0001$; paired t test), but the increase was greater for the intervention group ($p = 0.03$) (Table II). Greater kcal/week for the intervention group were also found for the 4-month difference in total physical activity (Tables II and III).

Secondary Outcomes

Follow-up Paffenbarger Index values were obtained for the 8-month outcome (89 patients in the control group and 104 patients in the intervention group) and the 12-month outcome (122 patients in the control group and 110 patients in the intervention group), resulting in 92% retention. The final mean follow-up time was 12.7 ± 1.3 months for the control group and 12.5 ± 1.2 months for the intervention group. Longitudinal results demonstrate that the initial 4-month walking and total activity gains were sustained during the longer-term follow-up period (Fig. 2). However, 95% confidence intervals overlapped for all values, indicating that there were wide variations within groups.

Throughout the trial, all patients reported life events that occurred since enrollment that they perceived impeded physical activity, including another lumbar surgery in 30 patients (12%) and lower-extremity surgery (hip and knee arthroplasty or foot or ankle surgery) in 74 patients (30%). In multivariable models, the intervention remained associated with greater kcal/week, and the reported events were associated with fewer kcal/week for the 12-month walking and total activity outcomes (Table III).

The pedometer was used to provide physiological feedback of measurable change. At the 12-month follow-up, 59% of patients in the intervention group reported that the pedometer was helpful at the beginning of the trial, 16% reported that it was helpful throughout the trial, and 25% reported that it was not helpful.

Adverse Events

At 12 months, 66% of patients in the intervention group and 78% of patients in the control group reported that being in the trial had no effect on the spine symptoms, and 34% of patients in the intervention group and 22% of patients in the control group reported that being in the trial made the spine symptoms better (odds ratio [OR], 1.8 [95% confidence interval (CI), 1.0 to 3.2]; $p = 0.05$). There were no trial-related adverse events reported in either group.

Discussion

A multicomponent behavioral intervention to increase physical activity after initial recuperation from complex lumbar surgery was effective in increasing kcal/week from walking and from total activities after 4 months, and the gain was sustained at 12 months. These gains were achieved without worsening spine symptoms. This trial builds on our prior work identifying patient-reported barriers to physical activity and offers an effective intervention based on established behavioral principles^{5,6}.

Our trial can impact patient care in several ways. First, our results highlight that walking more, compared with formal exercise, is a practical behavior that patients willingly accept. Second, fostering physical activity should not be restricted to primary-care settings but should be integrated into spine surgery practices where maximizing future spine health is a priority. Third, surgeons should be aware that subsequent surgical procedures can derail patients from physical activity; thus, reinforcing physical activity should be part of routine follow-up care.

We enrolled patients several months postoperatively when they had recuperated sufficiently and would potentially be motivated to resume active lifestyles (i.e., at a teachable moment when individuals are more receptive to behavior change)²⁹. Approval from surgeons provided confirmation not only that it was safe to increase activity but also that it was advisable. This type of approbation cannot be duplicated by other health-care providers who are not knowledgeable about the nature and/or extent of surgery and may be reluctant to persuade patients to increase activity. Thus, our trial supports including what otherwise would be a general health intervention into surgical practice.

We found wide variations in activity levels throughout the trial for both groups. These variations likely reflected other variables associated with inactivity that we did not account for, such as availability of places to walk, the family's fear of injury, and seasonal variations. Future studies should address these issues as well as account for the history of physical activity, include longer follow-ups, and include new motion-tracking devices and technologies that patients prefer (i.e., other than a pedometer). Future studies also could ascertain which components of the intervention are most effective for this population and clinical scenario.

We found peak activity levels at around 4 months and then modest declines. This finding is consistent with the well-known challenges of sustaining activity gains and the possible

need for a booster to reinforce behavior¹⁵. In future studies, such boosters could be administered with reapplication of the initial intervention. This would be pertinent in our sample because intervening life events, such as other surgeries impacting mobility, were prevalent.

Previous reports considering physical activity after lumbar surgery were mostly observational and did not include behavioral interventions. Some studies assessed self-reported walking difficulty and performance-based measures, such as gait speed, and reported improvements compared with preoperative status^{2,4,8,9,16,30}. These studies identified multiple covariates, including age, comorbidity, disability, body mass index, fear-avoidance, and surgical complexity^{2,3,7,8,31}. Among longitudinal studies, peak improvements were reported at 3 to 6 months postoperatively, followed by sustained improvements or moderate decreases at 12 months^{8,16}.

Our study had several limitations that could be addressed in future research. First, this study was conducted at a single urban Northeast tertiary-care center with primarily White patients and may not be applicable to other patients in different settings. Second, to administer the intervention longitudinally and maintain rapport with patients, study personnel were not blinded to the randomization group. This strategy did result in excellent retention rates. Third, we relied on a self-reported scale to measure activity. However, this scale is well-established and has been widely used in epidemiologic studies of physical activity. Fourth, although patients were given the same rehabilitation protocol, we did not quantify the duration or intensity of postoperative physical therapy. Fifth, we did not track other activity-related outcomes such as weight.

In summary, in this randomized controlled trial of patients enrolled 3 months after complex lumbar surgery, a multicomponent behavioral intervention was effective in increasing physical activity, primarily lifestyle walking, 4 months after enrollment and benefits were sustained to 12 months. These findings support counseling patients during routine postoperative care to increase physical activity in order to enhance both general health and spine health. ■

Carol A. Mancuso, MD^{1,2}
Roland Duculan, MD¹
Manuela C. Lafage, BSN³
Frank P. Cammisa, MD¹
Andrew A. Sama, MD¹
Alexander P. Hughes, MD¹
Darren R. Lebl, MD¹
Federico P. Girardi, MD¹

¹Hospital for Special Surgery, New York, NY

²Weill Cornell Medical College, New York, NY

³New York University Langone Health, New York, NY

Email for corresponding author: mancusoc@hss.edu

References

1. U.S. Department of Health and Human Services. 2008 Physical Activity Guidelines for Americans Accessed 2024 Sep 10. https://health.gov/paguidelines/second-edition/pdf/Physical_Activity_Guidelines_2nd_edition.pdf
2. McIlroy S, Jadhakhan F, Bell D, Rushton A. Prediction of walking ability following posterior decompression for lumbar spinal stenosis. *Eur Spine J*. 2021 Nov;30(11):3307-18.
3. Jakobsson M, Brisby H, Gutke A, Hägg O, Lotzke H, Smeets R, Lundberg M. Prediction of objectively measured physical activity and self-reported disability following lumbar fusion surgery. *World Neurosurg*. 2019 Jan;121:e77-88.
4. Budithi S, Dhawan R, Cattell A, Balain B, Jaffray D. Only walking matters: assessment following lumbar stenosis decompression. *Eur Spine J*. 2017 Feb;26(2):481-7.
5. Mancuso CA, Duculan R, Girardi FP. Healthy physical activity levels below recommended thresholds two years after lumbar spine surgery. *Spine (Phila Pa 1976)*. 2017 Feb 15;42(4):E241-7.
6. Mancuso CA, Rigaud MC, Wellington B, Duculan R, Cammisa FP, Sama AA, Hughes AP, Girardi FP. Qualitative assessment of patients' perspectives and willingness to improve healthy lifestyle physical activity after lumbar surgery. *Eur Spine J*. 2021 Jan;30(1):200-7.
7. McIlroy S, Walsh E, Sothinathan C, Stovold E, Norwitz D, Norton S, Weinman J, Bearne L. Pre-operative prognostic factors for walking capacity after surgery for lumbar spinal stenosis: a systematic review. *Age Ageing*. 2021 Sep 11;50(5):1529-45.
8. Quek JMT, Tan J, Toh I, Chen JL, Yeo W, Chia R, Ang A, Pua YH. Factors associated with pain intensity and walking disability after lumbar fusion: a longitudinal study. *Spine (Phila Pa 1976)*. 2022 Apr 15;47(8):597-606.
9. Takenaka H, Sugiura H, Kamiya M, Nishihama K, Ito A, Suzuki J, Kawamura M, Hanamura S, Hanamura H. Predictors of walking ability after surgery for lumbar spinal canal stenosis: a prospective study. *Spine J*. 2019 Nov;19(11):1824-31.
10. Skou ST, Pedersen BK, Abbott JH, Patterson B, Barton C. Physical activity and exercise therapy benefit more than just symptoms and impairments in people with hip and knee osteoarthritis. *J Orthop Sports Phys Ther*. 2018 Jun;48(6):439-47.
11. Bandura A. *Social Foundations of Thought and Action: A Social Cognitive Theory*. Englewood Cliffs, NJ: Prentice-Hall; 1986.
12. Sallis JF, Haskell WL, Fortmann SP, Vranizan KM, Taylor CB, Solomon DS. Predictors of adoption and maintenance of physical activity in a community sample. *Prev Med*. 1986 Jul;15(4):331-41.
13. Epstein LH, Wing RR, Thompson JK, Griffin W. Attendance and fitness in aerobics exercise: the effects of contract and lottery procedures. *Behav Modif*. 1980;4:465-79.
14. Anderson ES, Wojcik JR, Winett RA, Williams DM. Social-cognitive determinants of physical activity: the influence of social support, self-efficacy, outcome expectations, and self-regulation among participants in a church-based health promotion study. *Health Psychol*. 2006 Jul;25(4):510-20.
15. Mancuso CA, Choi TN, Westermann H, Wenderoth S, Hollenberg JP, Wells MT, Isen AM, Jobe JB, Allegrante JP, Charlson ME. Increasing physical activity in patients with asthma through positive affect and self-affirmation: a randomized trial. *Arch Intern Med*. 2012 Feb 27;172(4):337-43.
16. Takenaka H, Kamiya M, Sugiura H, Nishihama K, Suzuki J, Hanamura S. Recovery of the Japanese Orthopedic Association Back Pain Evaluation Questionnaire score and walking ability following lumbar spinal stenosis surgery. *Eur Spine J*. 2024 Jun;33(6):2222-33.
17. Mirza SK, Deyo RA, Heagerty PJ, Konodi MA, Lee LA, Turner JA, Goodkin R. Development of an index to characterize the "invasiveness" of spine surgery: validation by comparison to blood loss and operative time. *Spine (Phila Pa 1976)*. 2008 Nov 15;33(24):2651-61, discussion 2662.
18. Thompson WR, Gordon NF, Pescatello LS. *American College of Sports Medicine. ACSM's Guidelines for Exercise Testing and Prescription*. Philadelphia: Lippincott Williams & Wilkins; 2010.
19. Paffenbarger RS Jr, Blair SN, Lee IM, Hyde RT. Measurement of physical activity to assess health effects in free-living populations. *Med Sci Sports Exerc*. 1993 Jan;25(1):60-70.
20. Ainsworth BE, Haskell WL, Herrmann SD, Meckes N, Bassett DR Jr, Tudor-Locke C, Greer JL, Vezina J, Whitt-Glover MC, Leon AS. 2011 Compendium of physical activities: a second update of codes and MET values. *Med Sci Sports Exerc*. 2011 Aug;43(8):1575-81.
21. Charlson ME, Pompei P, Ales KL, MacKenzie CR. A new method of classifying prognostic comorbidity in longitudinal studies: development and validation. *J Chronic Dis*. 1987;40(5):373-83.
22. Kroenke K, Strine TW, Spitzer RL, Williams JBW, Berry JT, Mokdad AH. The PHQ-8 as a measure of current depression in the general population. *J Affect Disord*. 2009 Apr;114(1-3):163-73.
23. Hudson-Cook N, Tones-Nicholson K, Breen ARM, Jenner JR. A Revised Oswestry Disability Questionnaire. In: *Back Pain: New Approaches to Rehabilitation and Education*. Manchester, U.K.: Manchester University Press; 1989. p 187-204.
24. Waddell G, Newton M, Henderson I, Somerville D, Main CJ. A Fear-Avoidance Beliefs Questionnaire (FABQ) and the role of fear-avoidance beliefs in chronic low back pain and disability. *Pain*. 1993 Feb;52(2):157-68.
25. Looiman TJM, Dontje ML, Sprenger WP, van der Schans CP, de Groot M. Reliability and validity of ten consumer activity trackers. *BMC Sports Sci Med Rehabil*. 2015;7:24.
26. Melanson EL, Knoll JR, Bell ML, Donahoo WT, Hill JO, Nysse LJ, Lanningham-Foster L, Peters JC, Levine JA. Commercially available pedometers: considerations for accurate step counting. *Prev Med*. 2004 Aug;39(2):361-8.
27. Leon AS, Connett J, Jacobs DR Jr, Rauramaa R. Leisure-time physical activity levels and risk of coronary heart disease and death. The Multiple Risk Factor Intervention Trial. *JAMA*. 1987 Nov 6;258(17):2388-95.
28. Sesso HD, Paffenbarger RS, Ha T, Lee IM. Physical activity and cardiovascular disease risk in middle-aged and older women. *Am J Epidemiol*. 1999 Aug 15;150(4):408-16.
29. Brust M, Gebhardt WA, Ter Hoeve N, Numans ME, Kieffe-de Jong JC. Exploring timing and delivery of lifestyle advice following an acute cardiac event hospitalization: the cardiac patient's perspective. *Patient Educ Couns*. 2024 Jul;124:108279.
30. Wada T, Tanishima S, Kitsuda Y, Osaki M, Nagashima H, Noma H, Hagino H. Walking speed is associated with postoperative pain catastrophizing in patients with lumbar spinal stenosis: a prospective observational study. *BMC Musculoskelet Disord*. 2022 Dec 20;23(1):1108.
31. Gilmore SJ, Hahne AJ, Davidson M, McClelland JA. Predictors of substantial improvement in physical function six months after lumbar surgery: is early post-operative walking important? A prospective cohort study. *BMC Musculoskelet Disord*. 2019 Sep 11;20(1):418.